A Low-Cost Real-time Multimodal Seizure Detection System Using MAX30102 and Accelerometry

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Abstract

Seizure detection systems that rely on one physiological signal alone have often been plagued with high false alarm scenarios and low reliability in dynamic real-world environments. This paper is a low-cost, real-time seizure monitoring system using cardiovascular and motion-based features for detecting seizures with improved false-positive reduction. The proposed system entails integration of MAX30102 based heart rate sensor, MPU6050 accelerometer and real-time clock, and controlled from a Arduino Uno platform. Monitoring and analysis of heart rate irregularities that occur below 60 BPM or above 100 BPM along with sudden intense motions within the body that exceed 350 units are consistently conducted. A seizure event is only detected when both abnormal motion and heart rate concomitant deviations occur, and hence false positives are reduced. In the event of its detection, the system creates an audiovisual alert that needs manual resetting, and at the same time records the occurrence in timestamp format. Experimental observations demonstrate reliable detection of abnormal physiological conditions with alert responsive mechanisms. The proposed design is portable, energy-efficient and cost-effective, which can be used as home-based healthcare design and assistive medical device use.

Keywords: Seizure detection; wearable health monitoring; heart rate variability; accelerometer-based motion analysis; embedded medical systems; real-time monitoring; physiological signal fusion; assistive healthcare technology

1 Introduction

Neurological disorders, particularly epilepsy, affect millions of individuals globally and frequently necessitate continuous monitoring to ensure patient safety and enable prompt intervention. Seizures are usually manifested by sudden physiological changes such as unusual heart rate patterns and involuntary movements of the body. In many instances seizures happen unpredictably and outside of clinical settings and constant monitoring in hospital-based settings is impractical. As a result there is an emerging need for portable, in-the-moment seizure detection systems that are able to reliably operate in day-to-day living conditions and instantly alert the caregivers.

Traditional seizure monitoring approaches rely heavily upon electroencephalography (EEG) which, while clinically effective, is expensive, intrusive and cannot be used for long-term ambulatory use. Recent research has therefore focused on wearable and embedded systems with non-invasive physiological signals such as heart rate, motion and oxygen saturation [1]. However, systems based on only one modality are prone to high false alarm rates that are caused by normal physical activity, stress or environmental factors, and are therefore limited in their usability in the real world.

To overcome these problems, multimodal seizure detection frameworks have attracted more attention. By integrating the signals from the cardiovascular and motion-based systems, it is possible to obtain complementary physiological responses for seizure events. Heart rate abnormalities such as sudden changes in heart rate to a low or high value often accompany epileptic episodes while violent or rhythmic movements of body parts are characteristic of motor seizures. The fusion of these signals makes it possible to achieve more robust signal detection because the proposed system reduces the ambiguity through the presence of isolated fluctuations in the signal.

Figure 1 shows the general layout of the proposed multimodal system of detecting seizures. The background of the system starts with the physiological data obtained, which includes the Max30102 heart rate sensor which is an optical sensor, the Mpu6050 sensor used to detect body motion and the real time clock DS3231 in order to synchronize time functionality. The Arduino Uno micro-controller tasks these sensor values in which the signal filtering, heart rate calculation, and magnitude of motion calculations are done. The processed signals are then compared to a set of pre-established physiological thresholds within the seizure detecting algorithm. Should an abnormal heart rate and abnormal movement occur at the same time, the system would mark the occurrence as a seizure and trigger the alert system containing a buzzer, alarm LED and a display notification. The alarm condition can only be disabled by the user in case of a manual reset, so that no seizure events can be omitted.

And unlike traditional monitoring strategies, which may need to be interpreted by a clinician or monitored by a user, the system is fully automatic, offering real-time detection without the need for human intervention, by relying solely on built-in signal processing and decision-making rules.

The proposed system is made as standalone, low cost and energy efficient suitable for home-based healthcare and assistive monitoring application. By taking advantage of off-the-shelf sensors and a Arduino Uno-based design, the system aids in keeping the amount of hardware down at a minimum, while ensuring the reliable real-time operation of the system.

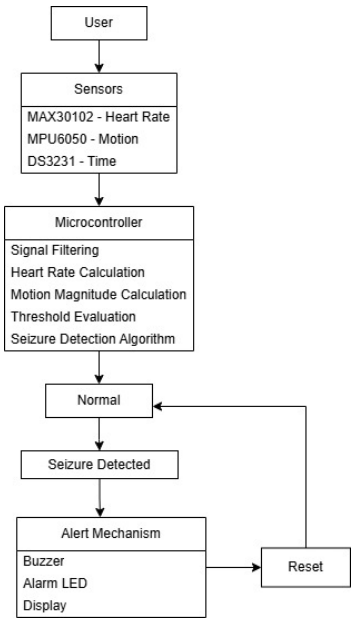


Fig. 1 System diagram of the seizure detection system.

- Methodology design of a multimodal seizure detection system
- A threshold-based fusion method for false alarm reduction
- An integrated alert system with LCD-based real-time status display indicating normal and seizure conditions

2 Literature Survey

Poh et al. [2] researched the possibility of non-EEG physiological signals to monitor seizures by employing autonomic nervous system responses to epileptic events. Their study showed that there were changes in cardiovascular-related signals that could be used to reliably indicate seizure activity, showing the feasibility for heart rate-based seizure detection. This work gives good motivation to use optical heart rate sensor in wearable and embedded seizure monitoring systems.

Van de Vel et al. [3] highlighted non-EEG seizure detection systems with attention to motion-based and cardiovascular sensing techniques. The importance of an accelerometer-based motion detection in the identification of convulsive seizures was emphasized by the authors, together with the possibility of combining various physiological signals to enhance the accuracy of detection. Their results provide support for the use of inertial sensors to provide real-time seizure monitoring in home-based environments.

Poh et al. [4] proposed a wearable multimodal seizure detection system that combined accelerometer signals and physiological signals to detect convulsive seizures. Their results showed the combined use of motion and physiological indicators was able to dramatically reduce false alarms, and improve upon single-sensor approaches. This

work is very much in line with multimodal fusion techniques used in embedded seizure detection systems.

Ulate-Campos et al. [5] presented a comprehensive evaluation of the automated seizure detection systems and discussed their practical limitations in the real world. The study pointed to some of the challenges, like false positives, sensor noise, and user comfort that need to be addressed so that robust algorithms for the detection and reliable alert mechanisms are presented. Their analysis subscribes to the need for multimodal sensing and alarm persistence in practical seizure monitoring solutions.

Beniczky et al. [6] discussed wearable seizure detection devices from the clinical usability perspective, concentrating on their use in continuous patient monitoring. The authors stressed the importance of in-time alerts and systems reliability for the human reaction of caregivers and safety for the patients. Their results highlight the importance of reliable alarm treatment and uninterrupted operation in seizure detection efforts in the home environment.

Jesper Jeppesen et al. [7] investigated seizure detection with heart rate variability (HRV) from wearable electrocardiography (ECG) monitors. A total of 100 patients were monitored with long-term video-EEG recordings, and 26 automatic models for HRV detection were tested against expert-annotated seizure events in a phase 2 study. The top performing model used indicators of sympathetic activity and rapid change in heart rate. It recognised 53.5% of patients as responders, with a sensitivity of 93.1% for all seizures and 90.5% for nonconvulsive seizures, 1.0 false alarms per 24 hours, and a median estimated latency to detection of 30 seconds. The researchers showed that HRV-based wearable devices are a practical and efficient method of seizure detection in patients with large autonomic changes.

Alamanda Hima Naga Rajasekhar et al. [8] developed an ensemble of models for the classification of epilepsy in multi-sequence MRI. Twelve deep learning models, such as CNN, MobileNet, InceptionV3, VGG, LSTM, and GRU, as well as combined models, were evaluated. The highest accuracy of 97.16%, achieved by the 'VGG19 + LSTM' ensemble, shows that this approach is the best at detecting epileptic abnormalities in MRI images. The findings show that the combination of spatial feature extraction (CNN) with temporal learning (LSTM/GRU) enhances classification performance, and ensemble models are a powerful approach for automating epilepsy detection systems.

Miglani and Ahuja [9] used the concept of deep learning, which combined CNN with GRU, as a solution for the automatic detection of epileptic seizures based on EEG signal data. Using the process of balancing the dataset by the use of spectrogram-based data preprocessing and augmentation, the model obtained an accuracy rate of 99.5%.

Raja Gowtham et al.[10] proposed a framework for extracting the maximum information from the most representative EEG channels and frequencies for seizure investigation. This framework involves standard preprocessing, wavelet decomposition, hierarchical segmentation, and peak-based event detection to extract relevant EEG channels and concentrate on low frequencies. The proposed work enhances computational efficiency and hence is appropriate for real-time and wearable EEG-based seizure detection.

Archana Reddy Penumada et al. [11] developed a real-time electrocardiogram (ECG) monitoring wearable system with IoT and machine learning for detecting heart

diseases. It continually collects ECG readings from wearable sensors and applies smart algorithms to detect early indicators of heart diseases. The IoT connection enables remote monitoring of patient health and real-time analysis and detection. This research demonstrates how the integration of wearable devices and machine learning offers a quick, non-invasive, and affordable approach to diagnosing heart diseases, enabling early detection and better preventive health care.

S. Nair et al. [12] investigated the advancement in epileptic seizure prediction using ECGs with deep learning. The study developed a patient alert system based on wearable ECGs instead of EEG, which is cumbersome to use in ambulatory settings. They implemented a deep neural network (DNN) baseline model with satisfactory performance. The authors aimed to improve the prediction accuracy by integrating convolutional neural networks (CNN) with signal decomposition methods, including empirical mode decomposition (EMD) and variational mode decomposition (VMD). The proposed method, which uses decision fusion of optimized features, substantially outperformed the baseline model, including improved accuracy (84.68%), sensitivity (69.60%), specificity (93.63%) and AUROC (0.8162). The findings show how deep learning approaches can be used in combination with signal processing techniques to enhance the accuracy of ECG-based seizure prediction systems, making it more effective for real-time healthcare.

3 Background and Methodology

3.1 Background

Seizure monitoring has traditionally been based on clinical-grade electroencephalography systems, which, while having high accuracy for diagnosis, cannot be used for continuous long-term monitoring at home or other locations outside a medical setting. This has led to an emphasis on research into non-invasive, wearable and embedded solutions that are able to function autonomously in real world environments.

Physiological signals that include heart rate and body motion have produced good correlations with seizure activity. Autonomic nervous system disturbance during seizures is a common result of sudden seizure that can affect the heart rate [13] while motor seizures are defined as sudden and intense involuntary movements. However, single-sensor approaches are prone to false triggers due to routine physical activity or environmental artifacts [14]. Consequently, multimodal sensing strategies that fuse complementary physiological indicators have become a promising solution that gives improved detection reliability.

3.2 System Architecture

The suggested seizure detection system adheres to a multimodal embedded architecture with physiological sensing, signal processing and alert generation enclosed in an autonomous system. Figure 2 indicates that the system is constructed based on Arduino Uno microcontroller connecting with three main modules that include the MAX30102 optical heart rate sensor, MPU6050 accelerator, and the DS3231 real-time clock (RTC). The MAX30102 reception of infrared photoplethysmography measures

to estimate the heart rate in beats per minute, and the MPU6050 measures tri-axis acceleration data utilized to determine the magnitude of motion indicating the intensity of the overall body movement. DS3231 RTC provides the detection of events and the successful stamping of these events to monitor and analyze. The Arduino Uno does real time signal processing, that is, one can filter the heart rate signal, compute the magnitude of motion movements based on acceleration elements, and physiological parameters based on a threshold value. To reduce the false alarms of single-sensor Fusion system, the seizure events are detected through a rule-based fusion algorithm, which assesses abnormal heart rate and abnormal motion states concurrently. As long as the both conditions are met, a system will trigger an alert system comprising of a buzzer, an LED indicator and a TFT LCD display aimed at alerting caregivers. The alarm state is held until manually cleared and therefore the seizure occurrences are not missed. This architecture allows a reliable, affordable and real-time seizure monitoring architecture that is appropriate in a home-based healthcare setting.

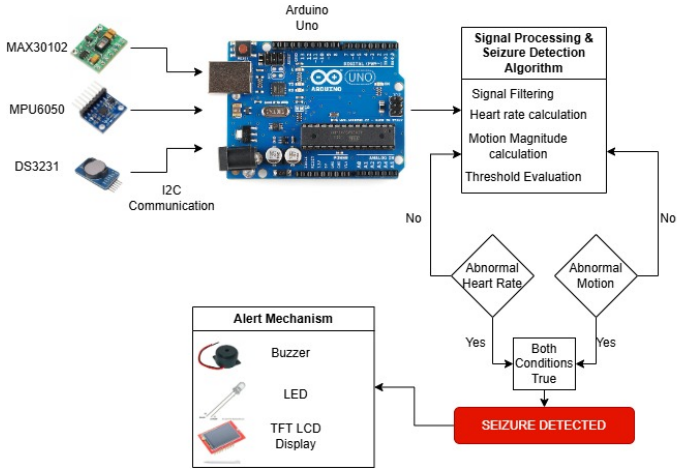


Fig. 2 System architecture of the proposed multimodal seizure detection system.

3.3 System Overview

The proposed system is an independent embedded seizure monitoring system consisting of a Arduino Uno, MAX30102-based heart rate sensor, MPU6050 accelerometer, real-time clock module, and TFT LCD display. The system continuously acquires physiological signals and processes these signals in real-time and identifies seizure events based on a rule-based fusion algorithm. Visual and auditory alerts are produced when a seizure is detected and the system will remain in alarm condition until manually cleared.

3.4 Data Acquisition

3.4.1 Heart Rate Signal Acquisition

Heart rate is obtained from an optical sensor that measures variations in the light absorbed by the body resulting from changes in blood volume. The infrared signal is sampled continuously and stored in a rolling buffer. A moving average filter is applied to reduce noise and stabilize the signal before performing beat detection. Only samples with adequate signal quality are taken into account for beat detection to avoid incorrect measurements.

3.4.2 Motion Signal Acquisition

Motion data are collected by a MPU6050 accelerometer with three orthogonal axes of acceleration measurements. The raw acceleration values are converted with respect to Earth gravity and combined to calculate a single motion magnitude to represent total body movement. This technique is sensitive to sudden and high-intensity movements that are indicative of seizure activity but is relatively insensitive to changes in orientation.

3.5 Signal Processing and Feature Extraction

3.5.1 Heart Rate Computation

Beat detection is done through the detection of peaks in the infrared signal that has been preprocessed. The distance between beats is used to compute the rate in terms of beats per minute (BPM). A heart rate of less than 60 BPM (bradycardia) and more than 100 BPM (tachycardia) is considered abnormal according to standard resting state physiology. The upper and lower limits are used to classify the heart rate value as normal or abnormal.

3.5.2 Motion Intensity Estimation

Motion intensity is assessed by computing the magnitude of the acceleration vector. This quantity is a scalar that is a brief indication of motion and is always compared with the threshold value to determine if there are any abnormalities of motion indicative of seizures. An acceleration greater than 350 units is considered abnormal, indicating sudden high-intensity motions not part of ordinary walking..

3.6 Seizure Detection Algorithm

The seizure detection logic is realized as a real-time, rule-based algorithm, which merges the signals from the cardiovascular and motion sensors in order to determine abnormal physiological events. The algorithm runs in the main execution loop of the embedded system and regularly evaluates the input sensor data. A seizure event is declared only when both abnormal motion intensity and abnormal heart rate conditions are satisfied at the same time, which reduces false alarms due to isolated signal fluctuations. Specifically, a heart rate outside the 60–100 BPM range combined with a motion magnitude greater than 350 units simultaneously triggers the alarm state.

Heart rate is calculated using the infrared MAX30102 data by detecting the inter-beat interval, and the motion intensity is estimated with the magnitude of MPU6050 acceleration data. Physiological threshold limits are used to categorize both signals as being either normal or abnormal. Once a seizure is detected, the system is put into a latched alarm state until it is manually reset by the user.

3.7 Algorithm (Pseudocode)

Algorithm 1 Multimodal Seizure Detection and Alert Handling

Input: Infrared signal from heart rate sensor, acceleration data from MPU6050 accelerometer
Output: Seizure alarm status (NORMAL / ALARM)
Initialize system parameters, thresholds, and alarm state
loop
    Acquire acceleration data along x, y, and z axes
    Compute motion magnitude from acceleration components
    Acquire infrared signal samples from heart rate sensor
    Apply rolling average filter to infrared signal
    Detect heart beats and compute heart rate (BPM)
    Classify heart rate as abnormal if $BPM < 60$ OR $BPM > 100$
    Classify motion as abnormal if motion magnitude > 350
    if motion is abnormal **and** heart rate is abnormal **then**
        Set alarm state to TRUE
    end if
    if alarm state is TRUE **then**
        Activate visual alert on display
        Activate buzzer and alarm LED
        Maintain alarm state (latched)
    end if
    if reset button is pressed **then**
        Clear alarm state
        Deactivate alerts
    end if
end loop

3.8 Implementation Details

The seizure detection system is a real-time embedded system based on the Arduino Uno microcontroller. The system includes a complete hardware configuration of a MAX30102 heart rate sensor, MPU6050 accelerometer, DS3231 real-time clock (RTC), 16x2 LCD, buzzer, LED indicators and a push button for manual reset. Figure 3 shows the physical hardware setup of the implemented system.

The sensor modules are connected to the Arduino Uno using the I2C communication protocol, allowing efficient and simultaneous data collection from multiple sensors

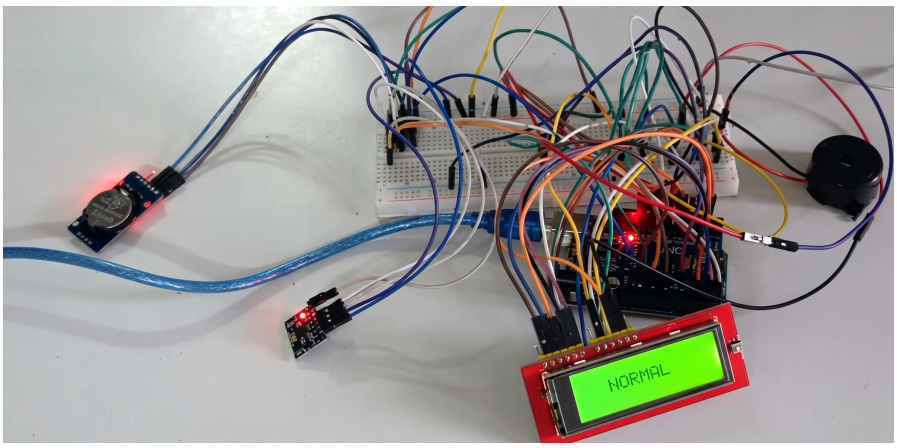


Fig. 3 Hardware implementation of the proposed seizure detection system showing Arduino Uno, sensor modules (MAX30102 and MPU6050), LCD display, and alert components interconnected via a breadboard setup.

in a bus-based system. The microcontroller operates in a polling loop to sample the sensors in real-time, eliminating the need for an additional processing unit.

The MAX30102 sensor module measures photoplethysmographic (PPG) signals, which are filtered on board with a moving average filter to eliminate noise and artefacts. A peak detection algorithm is applied to calculate the inter-beat interval, which translates into BPM (beats per minute) heart rate estimate. At the same time, the MPU6050 delivers tri-axial acceleration signals that are used to determine the motion magnitude, or the amount of body acceleration.

Our system uses a rule-based thresholding algorithm to determine whether the heart rate data and motion data are normal or abnormal. The system only raises an alarm for a seizure event when both criteria are met. This multimodal system reduces false alarms due to singlemode physiological variations.

When a seizure event is detected, the alarm is put in a latched mode. This involves the sounding of a buzzer, illumination of an LED and the display of a warning message on the LCD screen. The latched alarm condition continues until the system is manually reset via a push button.

The DS3231 real-time clock module is integrated to record the timing of events, allowing for event logging and possible clinical studies. The LCD interface also displays the real-time system status, showing NORMAL or SEIZURE DETECTED.

The entire system is self-contained, low-power, and does not require cloud-based processing or data analytics, thus being ideal for portable home care.

3.9 Warning and Handling of Alerts and System Control

For the system to guarantee the caregiver's awareness of the seizure events, the system uses an alarm latching mechanism. The system uses a push button for clearing the alarm. The system also uses a real-time clock for the seizure events. For the system's immediate awareness of the seizure events, the system uses a TFT LCD display.

4 Results

The performance test of the proposed system was conducted in several scenarios. The proposed system performance test scenarios include controlled and observational scenarios. The proposed system performance test scenarios are based on the system's operating conditions. The system output was monitored for the test.

4.1 Scenario 1: Normal Physiological Condition

The subject in this case was in a resting position or low activity but has normal heart rate and the amount of body movement was low. It is evidenced by the fact that the heart rate did not exceed the physiological range set, and the intensity of motion did not surpass the motion threshold. Within this set of circumstances, the system recognized the state as being normal during the period of observation. The TFT LCD screen showed normal status of the system and there was no visual or audible alerts evoked. The corresponding real-time monitoring data including stable heart rate and motion values is shown in Fig. 4. These observations prove that the system functions properly when it performs typical tasks and physiological functions (without false alarms) do not occur.

```
Time | Motion | BPM | State

Time: 2026-01-11T06:12:13 | Motion: 102.45 | BPM: 72.4 | State: NORMAL
Time: 2026-01-11T06:12:14 | Motion: 98.12  | BPM: 71.8 | State: NORMAL
Time: 2026-01-11T06:12:15 | Motion: 105.33 | BPM: 73.1 | State: NORMAL
Time: 2026-01-11T06:12:16 | Motion: 99.87  | BPM: 72.6 | State: NORMAL
Time: 2026-01-11T06:12:17 | Motion: 101.56 | BPM: 71.9 | State: NORMAL
```

Fig. 4 Real-time system output showing motion and heart rate under normal conditions.

4.2 Scenario 2: High Physical Activity without Heart Rate Abnormality

The scenario projected the actions like walking or light exercise that induce greater intensity of motion without uncharacteristic alterations of heart rate. Even with high motion rates registered by the accelerator the heart rate was in reasonable physiological bounds. The seizure alarm was not triggered as the heart rate was normal even though there was more movement of the body. The corresponding real-time monitoring output with high values of motion with normal heart rate and no seizure evidence is provided in Fig. 5. This finding is an indication of the efficiency of multimodal fusion strategy when compared to motion-only detection systems in false positive reduction.

```
Time: 2026-01-11T10:43:27 | Motion: 312.45 | BPM: 88.6 | State: NORMAL
Time: 2026-01-11T10:43:28 | Motion: 329.18 | BPM: 89.3 | State: NORMAL
Time: 2026-01-11T10:43:29 | Motion: 345.92 | BPM: 90.1 | State: NORMAL
Time: 2026-01-11T10:43:30 | Motion: 318.77 | BPM: 87.9 | State: NORMAL
Time: 2026-01-11T10:43:31 | Motion: 336.61 | BPM: 89.7 | State: NORMAL
```

Fig. 5 Real-time monitoring output showing high motion with normal heart rate and no seizure detection.

4.3 Scenario 3: Abnormal Heart Rate without Significant Motion

Abnormal heart rate pattern was recorded in such a situation, with minimal movement rates recorded of the body. These conditions could be either attributed to stress, anxiety, or transient sensor variations. Even though the system indicated the heart rate deviation, the magnitude of motion was not higher than the preset motion threshold. The abnormal motion condition was not met and as a result, the seizure alarm did not go off. The result of the corresponding real time monitoring output, showing abnormal heart rate values in low degree of motion with a normal system state, is given in Fig. 6. This phenomenon shows that the multimodal system proposed cannot detect false seizures because of isolated cardiovascular anomalies.

```
Time: 2026-01-11T14:19:42 | Motion: 96.42  | BPM: 132.6 | State: NORMAL
Time: 2026-01-11T14:19:43 | Motion: 101.25 | BPM: 135.2 | State: NORMAL
Time: 2026-01-11T14:19:44 | Motion: 99.84  | BPM: 130.9 | State: NORMAL
Time: 2026-01-11T14:19:45 | Motion: 103.11 | BPM: 133.8 | State: NORMAL
Time: 2026-01-11T14:19:46 | Motion: 97.56  | BPM: 134.5 | State: NORMAL
```

Fig. 6 Real-time monitoring output showing abnormal heart rate with normal system state.

4.4 Scenario 4: Simultaneous Abnormal Motion and Heart Rate (Seizure Event)

This artificial state was induced through the subject's deliberate attempt at fast and intense limb movements accompanied by physical activities (for example, vigorous exercise) that would increase the heart rate, mimicking – but not reproducing – the body's reaction to an epileptic seizure, which had no real seizures or patients data to refer to in this study. The two physiological indicators overcame their predefined thresholds concurrently, which activated the logic of seizure detection. When it was detected, the system was in alarm state where TFT LCD display showed that it had detected a seizure, the buzzer was turned on, and alarm LED stayed on. The respective real-time monitoring output, which depicts values of abnormal tmotion and the heart rate with the alarm state, is shown in Fig. 7. The alarm state was set to allow the event to be missed neither. The findings of these observations confirm the main detection property of the proposed multimodal seizure monitoring system.

```
Time: 2026-01-11T19:04:08 | Motion: 402.88 | BPM: 131.7 | State: ALARM
Time: 2026-01-11T19:04:09 | Motion: 428.53 | BPM: 138.4 | State: ALARM
Time: 2026-01-11T19:04:10 | Motion: 451.91 | BPM: 132.9 | State: ALARM
Time: 2026-01-11T19:04:11 | Motion: 439.67 | BPM: 130.2 | State: ALARM
Time: 2026-01-11T19:04:12 | Motion: 417.35 | BPM: 136.8 | State: ALARM
```

Fig. 7 Real-time monitoring output showing abnormal motion and heart rate with alarm state.

4.5 Scenario 5: Alarm Latching and Manual Reset

The reset button was actuated during Scenario 4 in an attempt to validate the alarm handling process following activation of the seizure alarm. The system maintained

an activated alarm status until the reset button was actuated, whereafter the alarm was deactivated instantly, the buzzer and LEDs switched off, and the display resumed to normal monitoring. The real-time monitoring output demonstrating alarm deactivation and restoration of the system is presented in Fig. 8. This proves the correct implementation of the alarm latching process and the effectiveness of the manual reset process.

```
Time: 2026-01-11T19:04:08 | Motion: 402.88 | BPM: 131.7 | State: ALARM
Time: 2026-01-11T19:04:09 | Motion: 428.53 | BPM: 138.4 | State: ALARM
Time: 2026-01-11T19:04:10 | Motion: 451.91 | BPM: 132.9 | State: ALARM
Time: 2026-01-11T19:04:11 | Motion: 439.67 | BPM: 130.2 | State: ALARM
Time: 2026-01-11T19:04:12 | Motion: 417.35 | BPM: 136.8 | State: ALARM
Alarm Reset → Cleared
Time: 2026-01-11T19:05:33 | Motion: 118.34 | BPM: 78.6 | State: ALARM
Time: 2026-01-11T19:05:34 | Motion: 110.92 | BPM: 76.9 | State: ALARM
Time: 2026-01-11T19:05:35 | Motion: 105.77 | BPM: 75.4 | State: ALARM
Time: 2026-01-11T19:05:36 | Motion: 101.63 | BPM: 74.8 | State: ALARM
Time: 2026-01-11T19:05:37 | Motion: 98.51  | BPM: 73.6 | State: ALARM
```

Fig. 8 Real-time monitoring output showing alarm reset and system recovery.

4.6 Scenario 6: Long-Term Continuous Operation

The system was put in the mode of continuous operation to test the stability and responsiveness of the system during long durations of monitoring. Real time sensor data was obtained and processed with no crashes or lost updates on the system. Logs that were time stamped ensured that things operated smoothly and that the physiological data were processed well in time. Fig. 9 shows the real-time output of monitoring performance under extended normal operating conditions. These findings prove that the suggested system has the potential to operate reliably over a long period and are applicable to the constant home-based healthcare monitoring.

```
Time: 2026-01-11T22:31:56 | Motion: 107.42 | BPM: 70.9 | State: NORMAL
Time: 2026-01-11T22:31:57 | Motion: 104.18 | BPM: 71.2 | State: NORMAL
Time: 2026-01-11T22:31:58 | Motion: 109.73 | BPM: 72.0 | State: NORMAL
Time: 2026-01-11T22:31:59 | Motion: 102.66 | BPM: 71.5 | State: NORMAL
Time: 2026-01-11T22:32:00 | Motion: 106.91 | BPM: 70.8 | State: NORMAL
```

Fig. 9 Real-time monitoring output during long-term normal operation.

5 Conclusion

This paper proposed a real-time embedded seizure monitoring system that incorporates the use of the MAX30102 based heart rate sensor and the MPU6050 accelerator based sensor of motion intensity to enhance reliability of seizure detection under non-clinical settings. The proposed system combines cardiovascular and motion sensors to overcome the shortcomings of single sensors, especially the elevated false alarm rates in the cases of normal physical exercises or the sole physiological oscillations.

The system has been installed on a low-cost Arduino Uno platform and tested in different realistic operation conditions. It was experimentally observed that the detection of correct seizures under the condition of abnormal heart rate and abnormal

motion co-occurring was achieved whereas normal patterns were not detected as false alarms. Improved awareness of the caregivers and reliability of the system was achieved by the latching alarm mechanism ensuring that detected events were visible until acknowledged by humans.

The offered design is self-sufficient, low-energy consuming, and offers visual and audible alerts, which would be beneficial in case of constant monitoring at home and assistive healthcare services. However, it is essential to point out that the system developed through this project is just a prototype meant for academic purposes and has not undergone any clinical validation processes it is not recommended to be used in practice until it is fully tested and approved. Although this approach uses fixed thresholding, which allows computational simplicity and inexpensive installation in order to minimize the burden on resources, the use of static thresholds does not allow adapting to the baseline of the individual patient, age-related physiological differences, and other diseases affecting the heart rate and body movement. Next iterations of this work will focus on the transition from static rules to more complex analytical approach of predictive analytics. Using machine learning classification algorithms [15] (such as lightweight decision trees, support vector machines, and recurrent neural networks like LSTM and GRU) on the basis of labeled multimodal signal data will allow replacing fixed thresholds with personalized decision boundaries, which will provide better sensitivity to different presentations of seizures and eliminate the false-positive alarms due to patient-specific variability. Since the problem is safety-critical, the new model should implement explainable AI (XAI). This means the use of feature attribution approaches (SHAP, etc.) or attention mechanisms in the architecture to make it possible to understand the reasons for generating alarms. Other promising areas include adaptive thresholding based on individual patient physiology, wireless transmission capabilities, and clinical validation of results using real datasets.

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